Bangladesh University of
Business and Technology (BUBT)

PROJECT PROPOSAL

on

Pharmacy Inventory Management System

Submission Date: **June 18, 2026**

Submitted by:

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01. NEEDS/PROBLEMS

Most small and medium-sized pharmacies in Bangladesh still manage their medicine stock and sales records manually. A pharmacist or sales staff typically keeps track of available medicines using paper registers or simple spreadsheet files. This approach causes several recurring problems:

- Stock records are hard to maintain manually. When a medicine is sold, the stock count is often not updated right away, which leads to confusion about what is actually available.
- There is no easy way to know which medicines are running low until they are completely out of stock. By then, customers may have already been turned away.
- Sales transactions are recorded by hand, which takes time and introduces errors. Finding a particular sale from past records becomes difficult.
- There is no access control in a manual system. Any employee can change stock or pricing data without any record of who made the change.
- Finding a specific medicine quickly when a customer is waiting is difficult when records are in a paper register or an unorganized file.
- Generating any kind of report, such as which medicines sold the most or which are near expiry, requires significant manual effort.

This project aims to solve these problems by building a simple, offline, console-based system that automates the core inventory and sales tasks of a pharmacy, while keeping a controlled access model so that sensitive operations require proper authorization.

02. GOALS/OBJECTIVES

The Pharmacy Inventory Management System is a console-based application developed in C. The aim is to build a working system that a small pharmacy can use to manage their daily medicine inventory and sales operations without needing internet access or any paid software.

The specific objectives of this project are:

- Build a role-based login system that separates Admin and Staff access, so that each user only gets the features they are authorized to use.
- Allow the Admin to add, update, search, and delete medicine records, while Staff members can perform these sensitive operations only after entering the admin password.
- Track stock quantities for each medicine and automatically reduce stock whenever a sale is made.
- Prevent sales from going through if a medicine is out of stock, so the system always reflects accurate availability.
- Alert the user whenever a medicine's stock falls below 10 units, giving enough time to reorder before it runs out.
- Record every sale with a unique Transaction ID, so individual transactions can be traced if needed.
- Allow both Admin and Staff to generate inventory, sales history, and low stock reports from within the system.

• Store all data in binary .dat files using structured file handling in C, without requiring any external database software.

03. EXISTING SYSTEM ANALYSIS

Before designing this system, we looked at a few existing pharmacy-related software solutions to understand what features they provide and where they fall short for small pharmacies in Bangladesh.

System	Type	Key Limitations Relevant to Our Project
Local Pharmacy POS Tools	Basic billing software (Excel / Windows Forms)	Handles billing only. No inventory integration, no role-based access control, no low stock alerts. Stock must be tracked separately. Files are prone to accidental corruption or deletion.
Medi-Soft	GUI-based pharmacy management software	Requires a paid license, making it unaffordable for small pharmacies. Heavyweight GUI with specific hardware requirements. Contains many features irrelevant to typical Bangladeshi pharmacy operations.

None of the above tools provide a combined solution for offline inventory management, role-based access control, low stock alerts, and sales tracking in a single system. Our system addresses these gaps by providing a lightweight, offline, console-based solution with proper role-based access control, automatic stock tracking, low stock alerts, and basic reporting, all within a single program that runs on any Windows PC with GCC installed.

04. FEATURES / SCOPES

The following is a realistic list of features that will be delivered in this project within the semester. No extra features beyond this list are planned.

04.1. Login System

- Admin login using username and password stored in a .dat file.
- Staff login using a separate username and password stored in the same file.
- After login, the system displays a role-specific menu.
- Three failed login attempts will exit the program.

04.2. Medicine Management

- Add medicine: Admin can add a new medicine record (ID, name, category, price, quantity). Staff can do this only after entering the Admin password.
- View all medicines: Both Admin and Staff can view the full list of medicines.
- Search medicine: Both roles can search by medicine name or medicine ID.
- Update medicine details: Admin can update any field. Staff requires Admin password.
- Delete medicine: Admin only. Staff requires Admin password to delete.

04.3. Inventory / Stock Management

- Each medicine record includes a stock quantity field.
- Staff can update the stock quantity when new stock arrives.
- The system displays a low stock warning for any medicine with quantity below 10 units.

04.4. Sales Management

- Both Admin and Staff can sell medicine.
- Each sale generates a unique Transaction ID in the format TXN001, TXN002, and so on.
- Each sale record stores: Transaction ID, Medicine ID, medicine name, quantity sold, unit price, total amount, date of sale, and username of the seller.
- The system automatically deducts the sold quantity from the medicine's stock.
- If the requested quantity is more than available stock, the sale is blocked and a message is shown.

04.5. Reports

- Inventory report: Displays all medicines with their current stock levels.
- Sales history report: Displays all past sale records with transaction details.
- Low stock report: Displays only medicines with stock below 10 units.

05. DEVELOPMENT MODEL

We have chosen the Incremental Development Model for this project.

In this model, the system is built in separate stages, where each stage adds a new working module to the existing system. Each module is designed, coded, and tested before the next one begins.

We chose this model for the following reasons:

- The project has five clearly separate modules: Login, Medicine Management, Inventory, Sales, and Reports. Each can be built and tested independently.
- If the team runs short on time near the end of the semester, we will still have a partially working and demonstrable system rather than nothing at all.
- It is easier to fix bugs when they appear within a small module than when the entire system is built at once.
- It allows team members to work on different modules simultaneously, which suits a 3-person team.
- Each completed increment can be shown to the supervisor for feedback before moving on.

The project will be built in four increments:

- Increment 01: Login system and file structure setup.
- Increment 02: Medicine management module (add, view, search, update, delete).
- Increment 03: Inventory tracking and Sales module.
- Increment 04: Reporting module and final integration and testing.

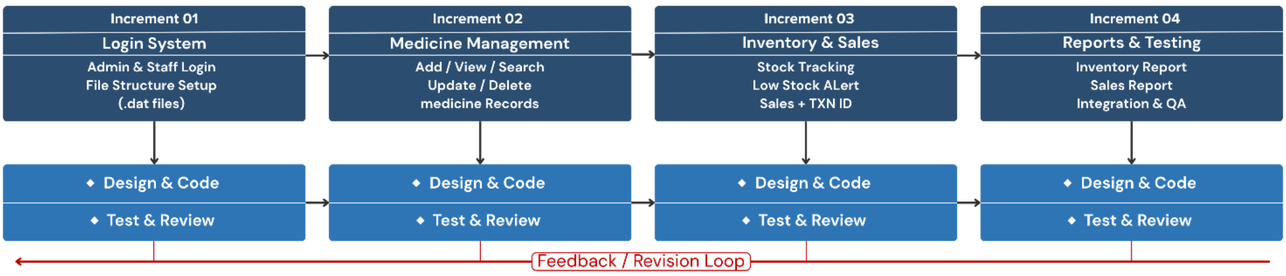

Figure 01: Incremental Development Model — Pharmacy Inventory Management System

06. FEASIBILITY ANALYSIS & POSSIBLE RISKS

06.1. Feasibility Analysis

Technical Feasibility

The entire system will be built using C programming language, file handling with .dat files, and standard I/O functions. All team members have already studied C programming and file handling in their coursework. The tools needed, GCC compiler and Visual Studio Code, are free and available on any Windows machine. No paid software or special hardware is required.

Economic Feasibility

There is no cost involved in this project. All tools are open source and free. The development machines used are the team members' personal laptops. No server, hosting, or software licensing is needed.

Operational Feasibility

The system is a console-based application that runs directly on any Windows PC. No installation is required beyond having a GCC compiler available. The target users, pharmacy Admin and Staff, need only basic computer literacy to operate the system. The menu-driven interface keeps the operations straightforward.

06.2. Possible Risks

- File corruption: If the program crashes while writing to a .dat file, the file could become corrupted. This risk will be reduced by keeping backups and handling file errors properly in the code.
- Scope creep: The team might be tempted to add extra features during development. To avoid this, the feature list in Section 4 is fixed and will not be changed unless approved by the supervisor.
- Team coordination issues: Since three members are working on different modules, there is a risk of naming conflicts or incompatible struct definitions. This will be managed by agreeing on all shared data structures before coding begins.
- Academic schedule conflicts: Exams and other coursework could reduce available development time. A buffer period is included in the time schedule for this reason.

07. REQUIREMENTS

Hardware Requirements

- Any PC or laptop with at least 2 GB RAM.

- At least 500 MB free disk space for the compiler, VS Code, and project files.
- Standard keyboard and monitor.

Software Requirements
- Operating System: Windows 10 or later (primary development environment). The system may also be tested on Linux using GCC, but Linux compatibility is not a project requirement.
- Compiler: GCC via MinGW-w64 or MSYS2 (Windows) / GCC via package manager (Linux).
- Editor: Visual Studio Code with C/C++ extension.
- Version Control: Git and GitHub for source code management.

Programming Language
- C (C11 standard).
- Standard C libraries: stdio.h, stdlib.h, string.h, time.h.

Data Storage
- All data will be stored in binary .dat files using C file handling (fread, fwrite, fseek).
- Separate .dat files for user accounts, medicine records, and sales transactions.

No Internet Required
- The system is fully offline. No network access is needed at any point.

07.1. Source Code Repository

The project source code will be maintained using GitHub for version control and team collaboration.

Repository Link: github.com/makhzumbinharun/pharmacy-inventory-system

GitHub will be used for:
- Source code management.
- Team collaboration.
- Progress tracking.
- Backup and version history.
- Documentation storage.

08. ACTIVITY AND TIME SCHEDULE/PROJECT BREAKDOWN WITH TIME ESTIMATION

#	Stages/ Tasks	Efforts, man-hours
Stage 1	**Analysis and Design**	
1.1	Requirements analysis and feature finalization	08
1.2	System design, module breakdown, and work plan	06

1.3	Data file structure design (.dat files for users, medicines, sales)	08
1.4	DFD and flowchart creation	06
Stage 2	**Implementation**	
2.1	Login module (Admin and Staff with file-based auth)	10
2.2	Medicine management module (add, view, search, update, delete)	20
2.3	Inventory and stock tracking module	15
2.4	Sales module with unique Transaction ID generation	15
2.5	Reports module (inventory, sales, low stock)	12
Stage 3	**Testing and other QA tasks**	
3.1	Unit testing for each module	10
3.2	Integration testing and bug fixing	08
Stage 4	**Deployment**	
4.1	Project report writing	06
4.2	Presentation and viva preparation	04
Total Estimated Efforts:		**128 man-hours**

09. BUDGET

This project does not require any financial investment. All tools, compilers, and software used are free and open source. The development is done on the team members' personal laptops and no external services are needed.

Project Item	Amount	Unit Price	Total
Software License (VS Code, GCC)	N/A	N/A	N/A (Free)
Development Machine(s)	N/A	N/A	N/A (Personal)
Database or Server License	N/A	N/A	N/A
Training Fees	N/A	N/A	N/A
Internet / Communication	N/A	N/A	N/A
Office Supplies / Printing	100	N/A	100
Total			BDT 100.00

10. DEVELOPMENT TEAM & STAKEHOLDERS

10.1. Development Team

Id	Name	Designation	Responsibilities
20254103279	MAKHZUM-BIN-HARUN	Team Leader & Core Programmer	Team coordination, system architecture design, Login module implementation, module integration

20254103269	MD. RAFIUL ISLAM	Core Programmer & SQA	Medicine Management module, file handling design, unit testing, test case documentation, bug tracking
20254103272	ESTIAK AHMED TURNAB	Core Programmer	Sales module, Inventory and stock tracking module, Reports module

10.2. Stakeholders

Stakeholder	**Role in System**	**Interest**
Pharmacy Owner	Admin	Wants full control over medicine records, stock, and business reports
Pharmacy Employee	Staff	Needs to handle daily sales and check stock without accessing sensitive management features
Course Supervisor	Academic Evaluator	Reviews project progress and evaluates the project for academic assessment.

11. LIMITATIONS & FUTURE WORKS

11.1. Limitations

The following are known limitations of this version of the system:

- The system supports only one Admin account and one Staff account. Multiple user accounts are not supported in this version.
- The application is single-user only. It does not support concurrent access from multiple machines or users at the same time.
- There is no graphical user interface. All interactions happen through the console/terminal.
- Expiry date tracking is not included in this version. Medicine expiry must be managed manually.
- Reports are displayed on the console screen only. There is no option to export or save a report to a file or print it.
- The system is primarily developed and tested on Windows. Limited testing may be performed on Linux. The system has not been tested on macOS, and compatibility with macOS is not guaranteed.

11.2. Future Works

The following features can be added in future versions if the system is extended:

- Support for multiple staff accounts with individual credentials and activity logs.
- Expiry date tracking with automatic alerts when a medicine is near expiry.
- Supplier and vendor management to track from whom medicines were purchased.
- Report export to a .txt or .csv file so records can be saved permanently or shared.
- A graphical user interface using a C GUI library or migration to another language.

- Barcode-based medicine search for faster lookup at the sales counter.

12. REFERENCES

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